

50 Lines of Python

Neighborhood DNA from Overture Maps Places

Any POI dataset will answer wrong twice.

Open data is the one kind that lets you catch it — in one screen of code.

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Where do we intervene first?

Before:

cross-country B2G research for EU policy institutions — 16 countries.
the functional layer meant licensed POI feeds or field surveys.

Every city client asks the same first question:

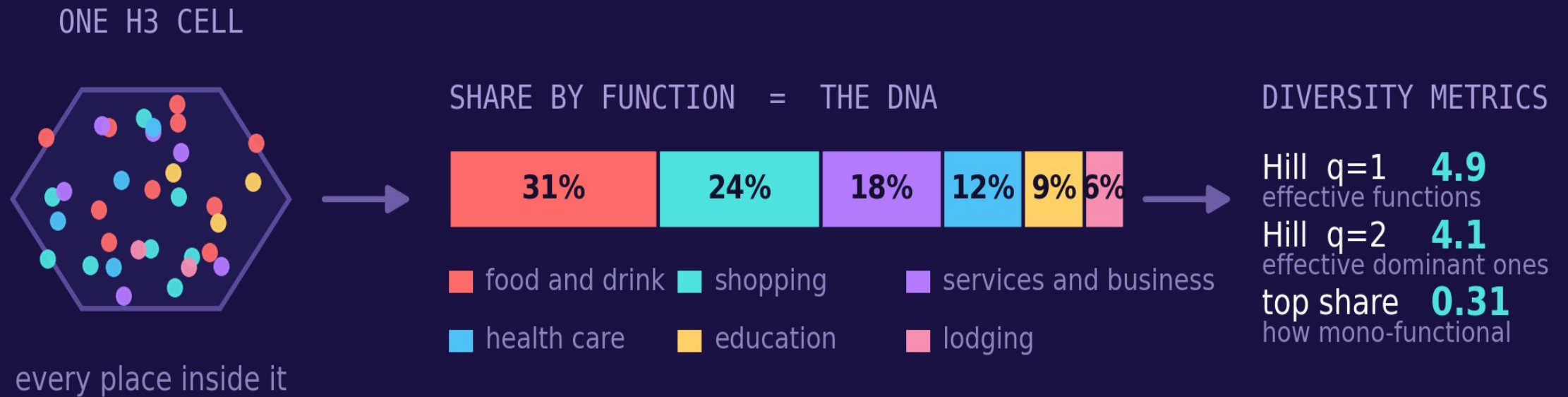
WHERE to start a mixed-use programme —
which blocks are so mono-functional they die after 19:00?

This talk:

the same shortlist from a fully open dataset.
Minutes on a laptop · 50 lines · reproducible by any municipal analyst.

Neighbourhood DNA — what it actually is

Take one grid cell. Look at every place inside it. Compute the share by function. That vector is the DNA — and from it, three numbers that say how mixed the cell is.



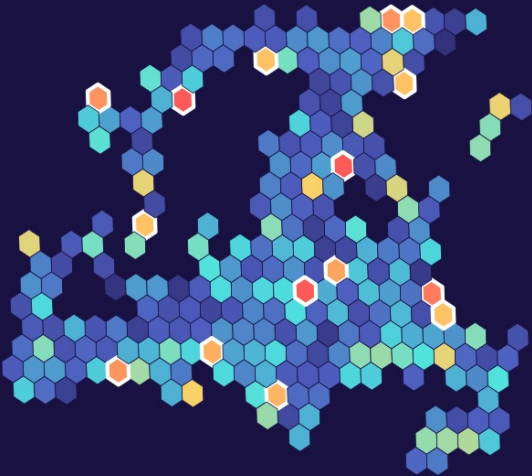
The same maths ecologists use for species in a habitat — applied to functions in a city block.

WHAT IT IS FOR

The blocks where one function owns the street

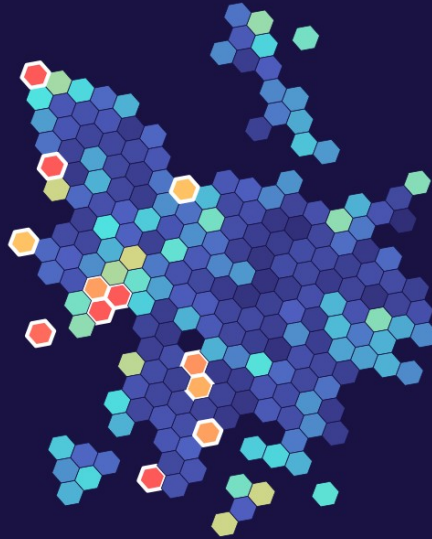
Hiroshima

15 flagged of 255



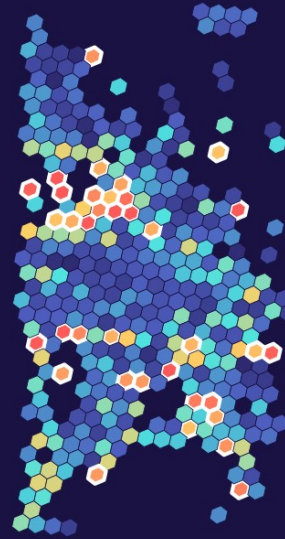
Belgrade

12 flagged of 221



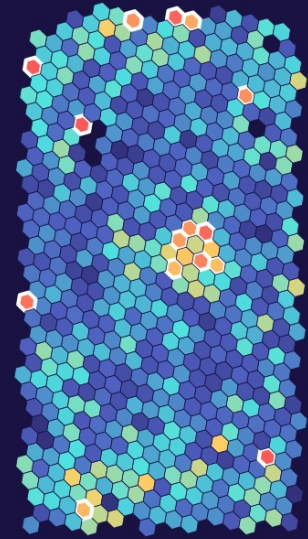
Amsterdam

36 flagged of 364



London

15 flagged of 586



top share — how mono-functional the cell is

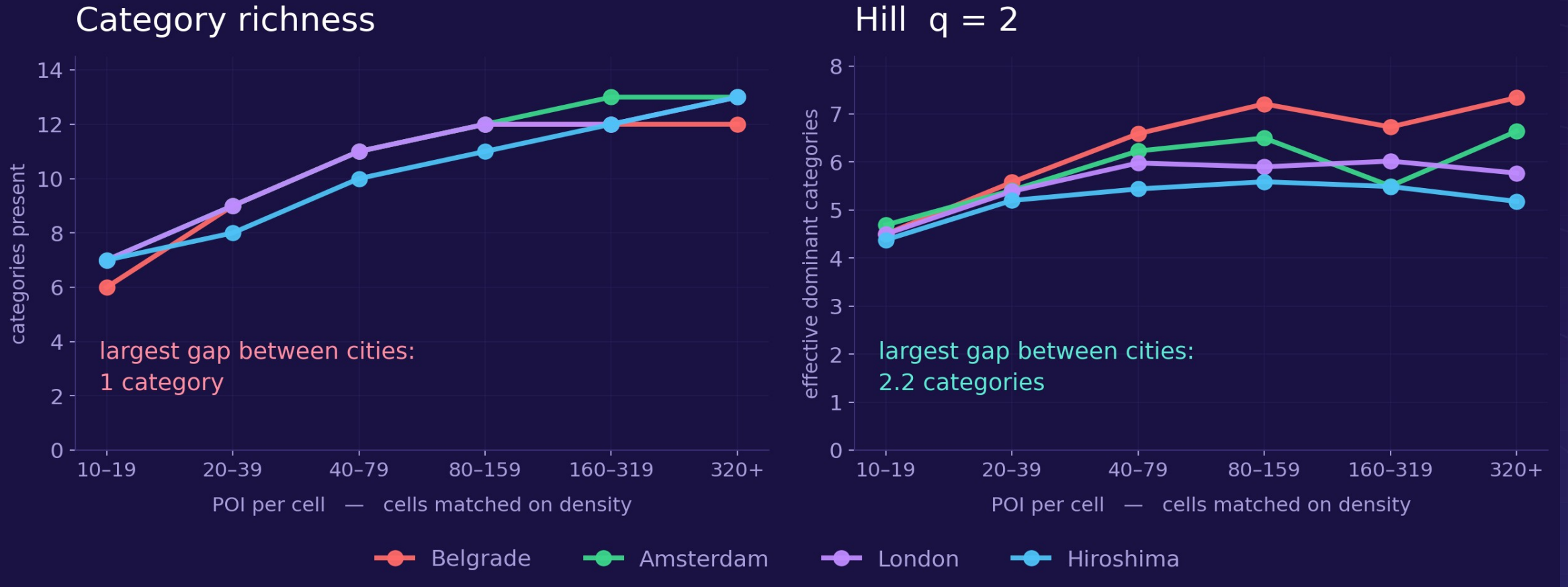


white outline = top share ≥ 0.5 , one function owns half the places · H3 res 8 · same 366 km² box in every city

Hexagons compare cities. Swap in your own blocks or districts to act on one.

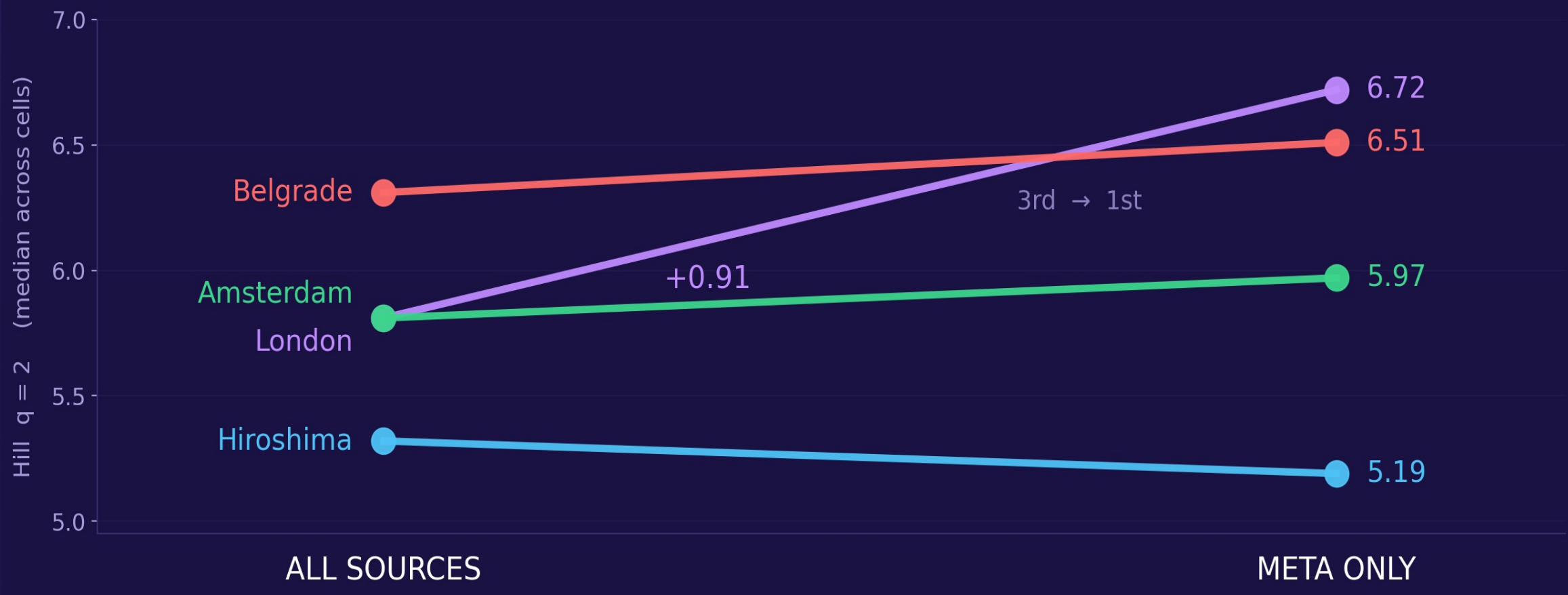
15 cells in Hiroshima, 12 in Belgrade. Short enough to walk. That is the shortlist planners never had.

The obvious metric sees nothing



Counting categories measures how much data you have. Hill numbers measure the city.

One filter flips the ranking



Overture blends providers. Belgrade is 93% one of them, London 62%. Nobody reads the sources column.

What Overture ships that makes this possible

GeoParquet on public S3

the bbox filter runs storage-side — no download, no ETL, no account

taxonomy.hierarchy

standardised hierarchy: level 1 = 13 branches to compare cities,
level 3 $\approx$ 200 leaves for detail

bbox struct in every row

predicate pushdown without touching geometry — the query returns in seconds

sources column

you can see whose POIs each city is — the basis of check #2;
closed feeds never show this

monthly releases + changelog

RELEASE pinned in code — July's numbers still reproduce in December

CDLA-Permissive 2.0

a municipal service without a legal department (ODbL layers: keep attribution)

No download. No ETL. One screen.

```
import sys, duckdb

RELEASE = "2026-07-22.0"
S3 = ("s3://overturemaps-us-west-2/release/"
      f"{RELEASE}/theme=places/type=place/*")
RES, MIN_POI = 8, 10
ymin, xmin, ymax, xmax, name = (
    *map(float, sys.argv[1:5]), sys.argv[5])

con = duckdb.connect()
con.sql("INSTALL httpfs; LOAD httpfs;")
con.sql("INSTALL h3 FROM community; LOAD h3;")
con.sql("SET s3_region='us-west-2';")

con.sql(f"""
CREATE TABLE cells AS
WITH places AS (
    SELECT h3_latlng_to_cell(
        bbox.ymin, bbox.xmin, {RES}) AS cell,
        taxonomy.hierarchy[1] AS category
    FROM read_parquet('{S3}',
        hive_partitioning = 1)
    WHERE bbox.xmin BETWEEN {xmin} AND {xmax}
        AND bbox.ymin BETWEEN {ymin} AND {ymax}
        AND taxonomy IS NOT NULL
),

shares AS (
    SELECT cell, category, n,
        n / sum(n) OVER (PARTITION BY cell) p
    FROM (SELECT cell, category,
        count(*)::DOUBLE n
        FROM places GROUP BY 1, 2)
)
SELECT cell,
    h3_cell_to_boundary_wkt(cell) AS geom,
    sum(n) AS poi_count,
    count(*) AS richness,
    -sum(p * ln(p)) AS shannon,
    exp(-sum(p * ln(p))) AS hill_q1,
    1 / sum(p * p) AS hill_q2,
    max(p) AS top_cat_share,
    arg_max(category, p) AS dominant
FROM shares
GROUP BY cell
HAVING sum(n) >= {MIN_POI};
""")

con.sql(f"COPY cells TO '{name}.parquet'")
```

DuckDB reads Overture straight off S3 — the bbox filter runs at the storage side. That is why it fits on one screen.

WHO USES THIS

Planners were just the example

Urban planning

Where to start a mixed-use programme

Transport

Where car dependency is built into the block

Retail siting

Which function is missing from a busy cell

Public health

Food deserts: homes with no grocery in reach

Tourism policy

Where lodging pushes out everyday functions

Humanitarian

Cities that have no other source at all

And one more: the same check that caught trap 2 measures bias in the open data itself.

One number — top share — answers a different question in every one of these rooms.

Two checks before you trust a POI metric

1 Control for cell size

Or you measure your data, not your city.

2 Check the sources column

Or you compare providers, not places.

Both checks are possible because the data is open — a paid feed never shows you its sources column.



Code, data and the full list of caveats

github.com/m-erts/foss4g2026

notebook: replace the city name, run, done · code MIT · slides CC BY 4.0

Overture release 2026-07-22.0 · the categories field is removed in the September release — write against taxonomy